

ORGANISATION – ENZYMES & DIGESTION

KEY VOCABULARY

Enzyme: a biological catalyst that speeds up reactions.

Substrate: the molecule an enzyme acts on.

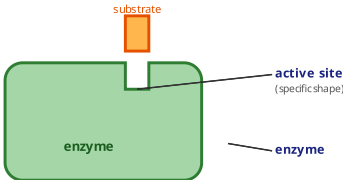
Active site: the part of the enzyme the substrate fits into.

Denatured: the active site changes shape and stops working.

COMMON MISCONCEPTIONS

- Enzymes are killed by high temperature.
- ✓ Enzymes are not alive — they are denatured.
- One enzyme works on any substrate.
- ✓ Enzymes are specific to one substrate shape.
- Denaturing can be reversed by cooling.
- ✓ Denaturing is permanent — the shape is lost.

LOCK-AND-KEY MODEL



Only a substrate with the right shape fits the **active site**.

1 DEFINE THE TERM

Describe what is meant by an **enzyme**. [2 marks]

2 LOCK AND KEY

Explain why each enzyme only works on one type of substrate. [2 marks]

3 EFFECT OF TEMPERATURE

Explain what happens to an enzyme above its optimum temperature. [3 marks]

4 DIGESTIVE ENZYMES

Complete the table of enzymes using the word bank. [2 marks]

___ breaks starch into ___; protease breaks protein into amino acids.

amino acids

glucose

protease

amylase

5 EFFECT OF PH

An enzyme works best at pH 7. Predict what happens to its rate at pH 3. [2 marks]

6 BILE

Explain how bile increases the rate of fat digestion. [2 marks]